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United States Patent	12388206
Kind Code	B2
Date of Patent	August 12, 2025
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High speed electrical connector for communication system

Abstract

An electrical connector includes a conductive housing having a mating end and a mounting end mounted to a circuit board. The conductive housing provides electrical shielding around pockets and includes ground pads at the mounting end configured to be terminated to the circuit board. The electrical connector includes contact assemblies held by the conductive housing each having a contact holder holding at least one contact. The contact holder is received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing. Each contact includes a mating end and a terminating end coupled to a signal pad of the circuit board. The conductive housing provides circumferential shielding around each contact assembly between the contact holder and the circuit board.

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Appl. No.: **17/941094**

Filed: **September 09, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240088587 A1	Mar. 14, 2024

Publication Classification

Int. Cl.: **H01R12/71** (20110101); **H01R12/51** (20110101); **H01R12/70** (20110101); **H01R13/631** (20060101); **H01R13/658** (20110101)

U.S. Cl.:

CPC **H01R12/716** (20130101); **H01R12/51** (20130101); **H01R12/7011** (20130101); **H01R13/631** (20130101); **H01R13/658** (20130101);

Field of Classification Search

CPC: H01R (12/76); H01R (12/51); H01R (12/7011); H01R (13/631); H01R (13/658)

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The subject matter herein relates generally to communication systems.

(2) There is an ongoing trend toward smaller, lighter, and higher performance communication components and higher density systems, such as for ethernet switches or other system components. Typically, the system includes an electronic package coupled to a circuit board, such as through a socket connector. Electrical signals are routed between the electronic package and the circuit board. The electrical signals are then routed along traces on the circuit board to another component, such

as a transceiver connector. The long electrical paths through the host circuit board reduce electrical performance of the system. Additionally, losses are experienced between the connector interfaces and along the electrical signal paths of the transceivers. Conventional systems are struggling with meeting signal and power output from the electronic package. Additionally, as data rates increase, conventional electrical connector designs suffer from signal integrity issues, such as at the interfaces between the electrical connector and the circuit board.

(3) A need remains for a reliable communication system.

BRIEF DESCRIPTION OF THE INVENTION

(4) In one embodiment, an electrical connector is provided and includes a conductive housing having a mating end and a mounting end. The mounting end configured to be mounted to a circuit board. The conductive housing includes pockets extending between the mating end and the mounting end. The conductive housing provides electrical shielding around the pockets. The conductive housing includes ground pads at the mounting end configured to be terminated to the circuit board. The electrical connector includes contact assemblies held by the conductive housing. Each contact assembly includes a contact holder holding at least one contact. The contact holder is received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing. Each contact includes a mating end and a terminating end. The terminating end configured to be coupled to a signal pad of the circuit board. The conductive housing provides circumferential shielding around each contact assembly between the contact holder and the circuit board.

(5) In another embodiment, an electrical connector is provided and includes a conductive housing having a mating end and a mounting end. The mounting end configured to be mounted to a circuit board. The conductive housing includes shielding walls extending between the mating end and the mounting end. The shielding walls define shield boxes surrounding pockets. The shield boxes provides electrical shielding on all sides of the pockets. The conductive housing includes ground pads along bottoms of the shielding walls configured to be terminated to the circuit board to electrically connect each shielding wall to the circuit board. The electrical connector includes contact assemblies held by the conductive housing. Each contact assembly includes a contact holder holding at least one contact. The contact holder is received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing. Each contact includes a mating end and a terminating end. The terminating end configured to be coupled to a signal pad of the circuit board. The shield boxes provide circumferential shielding around each contact assembly between the mating end and the mounting end of the conductive housing.

(6) In a further embodiment, a communication system is provided and includes a circuit board including an upper surface has a connector mounting footprint has an array of signal pads arranged in pairs. The signal pads are arranged in columns and rows. The circuit board includes a ground grid being electrically connected to a ground plane of the circuit board. The ground grid surrounds the pairs of signal pads. The ground grid includes ground strips extending along the columns and cross-strips extending along the rows. The cross-strips connect each of the ground strips. The communication system includes an electrical connector mounted to the connector mounting footprint. The electrical connector includes a conductive housing and contact assemblies held by the conductive housing. The conductive housing has a mating end and a mounting end mounted to the circuit board. The conductive housing includes pockets extending between the mating end and the mounting end. The conductive housing provides electrical shielding around the pockets. The conductive housing includes ground pads at the mounting end terminated to the ground grid of the circuit board. Each contact assembly includes a contact holder holding a pair of contacts. The contact holder is received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing. Each contact includes a mating end and a terminating end. The terminating end coupled to the corresponding signal pad of the circuit board.

The conductive housing provides circumferential shielding around each contact assembly between the contact holder and the ground grid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top view of a communication system having an electronic assembly in accordance with an exemplary embodiment.

(2) FIG. **2** is a top view of a portion of the circuit board in accordance with an exemplary embodiment showing one of the connector mounting footprints.

(3) FIG. **3** is a rear perspective view of a portion of the communication system showing one of the electrical connectors mounted to the circuit board and showing the cable connector module posed for coupling to the electrical connector in accordance with an exemplary embodiment.

(4) FIG. **4** is a front perspective view of a portion of the communication system showing one of the electrical connectors mounted to the circuit board and showing the cable connector module posed for coupling to the electrical connector in accordance with an exemplary embodiment.

(5) FIG. **5** is a top perspective view of the electrical connector in accordance with an exemplary embodiment mounted to the circuit board.

(6) FIG. **6** is a cross-sectional view of a portion of the communication system showing the cable connector module coupled to the electrical connector in accordance with an exemplary embodiment.

(7) FIG. **7** is a cross-sectional view of the electrical connector showing portions of the cable connector module coupled to the electrical connector in accordance with an exemplary embodiment.

(8) FIG. **8** is a bottom view of the electrical connector in accordance with an exemplary embodiment.

(9) FIG. **9** is a cross-sectional view of the electrical connector in accordance with an exemplary embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(10) FIG. **1** is a top view of a communication system **100** having an electronic assembly **102** in accordance with an exemplary embodiment. The electronic assembly **102** includes one or more cable connector modules **104** electrically connected to a circuit board **110** by one or more electrical connectors **200**. In various embodiments, the cable connector modules **104** may include optical modules using fiber optic cables for data transmission and/or electrical modules using electrical conductors and coper wire cables to transmit electrical data signals.

(11) In an exemplary embodiment, the communication system **100** includes an electronic package **106** electrically connected to the circuit board **110**. The cable connector modules **104** are electrically connected to the electronic package **106** through the circuit board **110**. In various embodiments, the electronic package **106** may be an integrated circuit assembly, such as an ASIC. However, the electronic package **106** may be another type of communication component. The electronic package **106** may be mounted to the circuit board **110**, such as being soldered to the circuit board **110**.

(12) In an exemplary embodiment, the electrical connectors **200** and the cable connector modules **104** are provided on multiple sides of the electronic package **106**, such as around the edge of the circuit board **110**. In the illustrated embodiment, the electrical connectors **200** and the cable connector modules **104** are provided on all four sides of the electronic package **106**. Other arrangements are possible in alternative embodiments. In the illustrated embodiment, twenty (20) electrical connectors **200** and corresponding cable connector modules **104** are coupled to the circuit board **110** around the electronic package **106**. Greater or fewer connectors may be provided in

alternative embodiments. Each electrical connector **200** is mounted to a connector mounting footprint **112** at a mounting area on an upper surface **114** of the circuit board **110**. The connector mounting footprint(s) **112** may be located adjacent the electronic package **106**. The electrical connectors **200** may additionally or alternatively be mounted to a lower surface of the circuit board **110**.

(13) FIG. **2** is a top view of a portion of the circuit board **110** in accordance with an exemplary embodiment showing one of the connector mounting footprints **112**. The circuit board **110** includes signal pads **120** and a ground grid **130** at the connector mounting footprint **112**.

(14) The signal pads **120** are arranged in an array, such as in rows and columns. The rows extend parallel to a primary axis **122** while the columns extend parallel to a secondary axis **124**. The signal pads **120** may be pads or traces of the circuit board **110**. In an exemplary embodiment, the signal pads **120** are high speed signal contacts. The signal pads **120** may be electrically connected to circuit traces routed through the circuit board **110**, such as on one or more layers of the circuit board **110**. In an exemplary embodiment, the signal pads **120** are arranged in pairs configured to transmit high speed, differential signals. In the illustrated embodiment, the connector mounting footprint **112** includes sixteen (16) pairs of the signal pads **120** arranged in a 4×4 array. Other arrangements are possible in alternative embodiments. Greater or fewer pairs of the signal pads **120** may be provided in alternative embodiments.

(15) The ground grid **130** surrounds the pairs of the signal pads **120**. The ground grid **130** defines, is part of, or is electrically connected to a ground plane of the circuit board **110**. The ground grid **130** may be electrically grounded. The ground grid **130** is separated from the signal pads **120**, such as by dielectric material and/or air. The ground grid **130** provides electrical shielding for the pairs of the signal pads **120**. In an exemplary embodiment, the ground grid **130** includes ground strips **132** and cross-strips **134** connecting the ground strips **132**. The ground strips **132** extend along the columns of the signal pads **120** and the cross-strips **134** extend along the rows of the signal pads **120**. For example, the ground strips **132** extend parallel to the primary axis **122** and the cross-strips **134** extend parallel to the secondary axis **124**. The ground strips **132** and the cross-strips **134** meet at intersections **136**. In an exemplary embodiment, the ground grid **130** is continuous, such as completely surrounding the pairs of signal pads **120** on all four sides of the pairs of signal pads **120**. For example, the ground grid **130** extends around the outer perimeter of the array of signal pads **120** and extends internally between the pairs of signal pads **120**. In various embodiments, the ground strips **132** and the cross-strips **134** may have the same widths. Alternatively, the ground strips **132** may be wider or narrower than the cross-strips **134**.

(16) FIG. **3** is a rear perspective view of a portion of the communication system **100** showing one of the electrical connectors **200** mounted to the circuit board **110** and showing the cable connector module **104** posed for coupling to the electrical connector **200**. FIG. **4** is a front perspective view of a portion of the communication system **100** showing one of the electrical connectors **200** mounted to the circuit board **110** and showing the cable connector module **104** posed for coupling to the electrical connector **200**. The cable connector module **104** is configured to be coupled to the electrical connector **200** from above.

(17) The cable connector module **104** includes a connector housing **150** (FIG. **3**) holding cable assemblies **152**. In an exemplary embodiment, the connector housing **150** includes a conductive insert **151** (FIG. **4**) and an outer shell **153** (FIG. **3**). The outer shell **153** may be manufactured from a dielectric material. Each cable assembly **152** includes a contact assembly **154** and a cable **156** terminated to the contact assembly **154**. The contact assembly **154** includes a contact holder **158** holding at least one contact **160** (shown in FIG. **4**), such as a pair of contacts, and a shield **162** providing shielding for the contact(s) **158**. The contact assemblies **154** may be arranged in an array, such as in rows and columns.

(18) With additional reference to FIG. **5**, which is a top perspective view of the electrical connector **200** in accordance with an exemplary embodiment mounted to the circuit board **110**. The electrical

connector **200** is mounted to the circuit board **110** at the connector mounting footprint **112**. The electrical connector **200** includes a conductive housing **210** holding a plurality of contact assemblies **240**. The contact assemblies **240** are electrically connected to the circuit board **110** and configured to be mated with corresponding contact assemblies of the cable connector module **104**. The conductive housing **210** provides electrical shielding for the contact assemblies **240**. The conductive housing **210** provides electrical shielding to the circuit board **110**. For example, the conductive housing **210** may be connected to the ground grid **130** of the circuit board **110**.

(19) The conductive housing **210** is manufactured from a conductive material, such as a plated plastic material that may be selectively plated, such as on all sides of each pocket. The conductive material may be a conductive polymer, a metallic compound, or a cast or molded metal material in alternative embodiments. The conductive housing **210** includes a mating end **212** and a mounting end **214**. The mating end **212** is configured to be mated to the cable connector module **104**. The mounting end **214** is configured to be mounted to the circuit board **110**. In the illustrated embodiment, the mating end **212** is at a top of the conductive housing **210** and the mounting end **214** is at a bottom of the conductive housing **210**. However, the mating end **212** may be at other orientations, such as at a right angle relative to the mounting end **214**. The conductive housing **210** includes sides **216** between the mating end **212** and the mounting end **214**. In various embodiments, the conductive housing **210** is box-shaped having four sides **216**. However, the conductive housing **210** may have other shapes in alternative embodiments.

(20) The conductive housing **210** includes pockets **220** extending between the mating end **212** and the mounting end **214**. The pockets **220** receive corresponding contact assemblies **240**. The conductive housing **210** provides electrical shielding around the pockets **220**. The pockets **220** may be open at the mating end **212** to receive portions of the contact assemblies of the cable connector module **104**.

(21) The conductive housing **210** includes ground pads **230** at the mounting end **214** configured to be terminated to the circuit board **110**. The ground pads **230** may be soldered to the circuit board **110**, such as to the ground grid **130**. The ground pads **230** are provided at the bottom of the conductive housing **210**.

(22) In an exemplary embodiment, each contact assembly **240** includes a contact holder **250** holding at least one contact **260**. In various embodiments, each contact holder **250** holds a pair of the contacts **260**, which define a differential pair. The contact holder **250** is received in the corresponding pocket **220** of the conductive housing **210** to position the contacts **260** relative to the conductive housing **210**. The contact holder **250** is manufactured from a dielectric material, such as a plastic material, to electrically isolate the contacts **260** from the conductive housing **210**. The conductive housing **210** provides shielding for the contacts **260**, such as between the mating end **212** and the mounting end **214**. In an exemplary embodiment, the conductive housing **210** provides circumferential shielding (for example, perimeter shielding) around each contact assembly **240**. The conductive housing **210** provides shielding to the circuit board **110**, such as to provide shielding at the interface between the contacts **260** and the circuit board **110**.

(23) In an exemplary embodiment, the conductive housing **210** includes shielding walls **222** forming the pockets **220**. The shielding walls **222** provide shielding on all sides of the pockets **220**. The shielding walls **222** include outer walls **224** around the outer perimeter of the conductive housing **210** and separating walls **226** within an interior of the conductive housing **210**. The separating walls **226** separate the pockets **220** from each other. The separating walls **226** provide electrical shielding between the pockets **220**. The outer walls **224** provide shielding around the exterior of the corresponding pockets **220**. In an exemplary embodiment, the pockets **220** are box shaped. However, the pockets **220** may have other shapes. The shielding walls **222** form shield boxes **228**. The shield boxes **228** provide shielding on four sides of each pocket **220**. In an exemplary embodiment, the pockets **220** are arranged in columns and in rows. The conductive housing **210** includes primary shielding walls **222*a*** extending along the columns of the pockets **220**

and secondary shielding walls **222***b* extending along the rows of the pockets **220**. The secondary shielding walls **222***b* extend between the primary shielding walls **222***a*. The primary shielding walls **222***a* extend parallel to the primary axis **122**. The secondary shielding walls **222***b* extend parallel to the secondary axis **124**. The pockets **220** in the rows are separated from other pockets **220** by the primary shielding walls **222***a*. The pockets **220** in the columns the separated from other pockets **220** by the secondary shielding walls **222***b*.

(24) FIG. **6** is a cross-sectional view of a portion of the communication system **100** showing the cable connector module **104** coupled to the electrical connector **200**. FIG. **6** shows one of the contact assemblies **154** coupled to the corresponding contact assembly **240** of the electrical connector **200**. The connector housing **150** and the conductive housing **210** create silos that surround and shield the contacts **160**, **260**. The shielding extends the lengths of the contacts **160**, **260**, such as from the circuit board **110** to the cable **156** (shown in FIG. **3**).

(25) In an exemplary embodiment, each contact holder **250** includes a base **252** and a tower **254** extending from the base **252**. The tower **254** may be integral with the base **252**, such as being co-molded with the base **252**. Alternatively, the tower **254** may be molded separate from the base **252** and coupled thereto. The base **252** includes a top **256** and a bottom **258**. In various embodiments, the base **252** is held in the pocket **220**, such as by an interference fit. The top **256** of the base **252** may be located below the top of the conductive housing **210**. The bottom **258** of the base **252** may be located above the bottom of the conductive housing **210**. The tower **254** extends from the top **256** of the base **252**, such as to a height above the mating end **212**. For example, the tower **254** may extend to the exterior of the conductive housing **210**, for example, into the connector housing **150**. The contact **260** extends along the tower **254**, such as into the connector housing **150**. The contact **260** extends from the bottom **258** of the base **252** to the circuit board **110**.

(26) In an exemplary embodiment, each contact **260** includes a mating end **262** and a terminating end **264**. The mating end **262** is configured to be mated to the contact **160**. The terminating end **264** is configured to be terminated to the circuit board **110**. In various embodiments, the mating end **262** includes a spring beam **266** having a mating interface. The spring beam **266** is deflectable. Other types of contacts may be provided in alternative embodiments, such as pins, sockets, blades, and the like. In various embodiments, the terminating end **264** includes a solder tail **268** configured to be soldered to the signal pad **120** of the circuit board **110**. The solder tail **268** is located below the bottom of the base **252** of the contact holder **250**. The solder tail **268** may be bent 90° to orient the solder tail **268** for soldering to the signal pad **120**. Other types of terminating ends may be provided in alternative embodiments, such as compliant pins.

(27) In an exemplary embodiment, the conductive housing **210** provides circumferential shielding around a portion of the contact **260** above the base **252**, such as the spring beam **266**. The conductive housing **210** provides circumferential shielding around a portion of the contact **260** below the bottom **258** of the base **252**, such as the solder tail **268**. In an exemplary embodiment, the conductive housing **210** provides shielding all the way from the base **252** to the circuit board **110**. For example, the ground pads **230** at the bottom of the conductive housing **210** are connected (for example, soldered) to the ground grid **130** of the circuit board **110**. As such, the solder tails **268** are entirely shielded to improve signal integrity at the interface with the circuit board **110**.

(28) FIG. **7** is a cross-sectional view of the electrical connector **200** showing portions of the cable connector module **104** coupled to the electrical connector **200**. When mated, the contact assemblies **154** are plugged into the pockets **220** of the conductive housing **210** to mated with the contact assemblies **240**. The shielding walls **222** form a grid between the pockets **220**. The shielding walls **222** provide shielding around the contact assemblies **154**, **240**. The shielding walls **222** provide electrical shielding between the pairs of the contacts **160**, **260**. The contact holders **158**, **250** electrically isolate the contacts **160**, **260** from the shielding walls **222**.

(29) In an exemplary embodiment, the shielding walls **222** provide circumferential shielding (for example, perimeter shielding) around the pockets **220**. The circumferential shielding follows the

shape of the perimeter of the pocket **220** (for example, may be rectangular, oval, circular or other shapes). In various embodiments, the circumferential shielding of the conductive housing **210** extends entirely circumferentially around each pocket **220** (for example, continuously or 360° or along all four sides). In other various embodiments, the circumferential shielding of the conductive housing **210** extends at least partially circumferentially around each pocket **220**. For example, the circumferential shielding may cover greater than 50% of the perimeter of the pocket **220**. The circumferential shielding may cover greater than 75% of the perimeter of the pocket **220** (for example, 270°). The circumferential shielding may be generally continuous, but may include small gaps or spaces, such as at slots in the shielding walls **22** that are provided for plugging of the contact assemblies **240** into the pockets **220**.

(30) FIG. **8** is a bottom view of the electrical connector **200** in accordance with an exemplary embodiment. The electrical connector **200** includes the conductive housing **210** holding the contact assemblies **240**. The conductive housing **210** provides electrical shielding for the contact assemblies **240**. The conductive housing **210** is configured to be connected to the ground grid **130** of the circuit board **110** (shown in FIG. **2**) to electrically common the conductive housing **210** with the ground plane of the circuit board **110**.

(31) The conductive housing **210** includes the ground pads **230** at the bottom of the conductive housing **210**. The ground pads **230** are defined by the bottoms of the shielding walls **222**. The ground pads **230** are configured to be coupled to the ground grid **130** of the circuit board **110**, such as being soldered or compression coupled to the ground grid **130**.

(32) The ground pads **230** form a grid **232** at the bottom that surrounds the terminating ends **264** of the contacts **260**. The ground pads **230** provide shielding on all sides of the pockets **220**. The ground pads **230** provide 360° shielding around the terminating ends **264** of the contacts **260**. In an exemplary embodiment, the ground pads **230** include ground pad strips **234** and ground pad cross-strips **236** connecting each of the ground pad strips **234**. The ground pad cross-strips **236** may be oriented perpendicular to the ground pad strips **234**. The separating walls **226** separate the pockets **220** from each other. The ground pad strips **234** and the ground pad cross-strips **236** provide electrical shielding between the pockets **220**. In an exemplary embodiment, the ground pad strips **234** extend along the columns of the pockets **220** and the ground pad cross-strips **236** extend along the rows of the pockets **220**. The ground pad strips **234** extend parallel to the primary axis **122**. The ground pad cross-strips **236** extend parallel to the secondary axis **124**. The pockets **220** in the rows are separated from other pockets **220** by the ground pad strips **234**. The pockets **220** in the columns the separated from other pockets **220** by the ground pad cross-strips **236**.

(33) In various embodiments, the shielding walls **222** include slots **238** at the bottom. The slots **238** form gaps between the ground pads **230**. The slots **238** are formed in the primary shielding walls **222*a***. The slots **238** are provided to receive the contact assemblies **240**. For example, the contact assemblies **240** may be plugged into the pockets **220** and extend across the slots **238** between the pockets **220**. As such, multiple contact assemblies **240** may be simultaneously loaded into the corresponding pockets **220**. The slots **238** are relatively small to have minimal impact on shielding at the bottom of the conductive housing **210**. For example, the slots **238** are narrow and the shielding walls **222** are capacitively coupled across the slots **238** to reduce EMI leakage through the slots **238**. As such, the conductive housing **210** maintains 360° shielding around the pockets **220**.

(34) FIG. **9** is a cross-sectional view of the electrical connector **200** in accordance with an exemplary embodiment. The shielding walls **222** form the pockets **220**, which receive the contact assemblies **240**. The base **252** of the contact holder **250** is positioned in the pocket **220** between the shielding walls **222**. The base **252** may be press fit in the pocket **220**. The solder tail **268** at the terminating end **264** of the contact **260** is located below the bottom of the base **252** for connection to the signal pad **120** of the circuit board **110**. The conductive housing **210** provides shielding below the bottom **258** of the base **252**, such as between the solder tails **268** in the adjacent pockets

222.
(35) In an exemplary embodiment, the conductive housing **210** provides shielding along the solder tails **268**, such as from the base **252** to the circuit board **110**. For example, the ground pads **230** are connected to the ground grid **130** of the circuit board **110**. As such, the solder tails **268** are entirely shielded to improve signal integrity at the interface with the circuit board **110**. In an exemplary embodiment, the ground pads **230** are coplanar with the terminating ends **264** of the contacts **260**. The conductive housing **210** provides circumferential shielding around the portions of the contacts **260** below the bottom **258** of the base **252**.
(36) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. An electrical connector comprising: a conductive housing having a mating end and a mounting end, the mounting end configured to be mounted to a circuit board, the conductive housing including pockets extending between the mating end and the mounting end, the conductive housing providing electrical shielding around the pockets, the conductive housing including ground pads at the mounting end configured to be terminated to the circuit board; contact assemblies held by the conductive housing, each contact assembly including a contact holder holding at least one contact, the contact holder being received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing, each contact includes a mating end and a terminating end, the terminating end configured to be coupled to a signal pad of the circuit board; wherein the ground pads are coplanar with the terminating ends of the contacts; and wherein the conductive housing provides circumferential shielding around each contact assembly between the contact holder and the circuit board.
2. The electrical connector of claim 1, wherein the conductive housing provides 360° shielding for each pocket.
3. The electrical connector of claim 1, wherein the conductive housing includes shielding walls forming the pockets, the shielding walls providing shielding on all sides of the pockets.
4. The electrical connector of claim 3, wherein the pockets are box shaped, the shielding walls forming shield boxes providing shielding on four sides of each pocket.
5. The electrical connector of claim 1, wherein the ground pads provide 360° shielding around the terminating ends of the contact.
6. The electrical connector of claim 1, wherein the ground pads are configured to be soldered to the circuit board.

7. The electrical connector of claim 1, wherein the ground pads form a grid including ground pad strips and ground pad cross-strips connecting each of the ground pad strips.
8. The electrical connector of claim 1, wherein the contacts are arranged in pairs, each pair of the contacts being shielded from adjacent pairs of the contacts by the conductive housing.
9. The electrical connector of claim 1, wherein the pockets are arranged in columns and in rows, the conductive housing having primary shielding walls extending along the columns of the pockets and secondary shielding walls along the rows of the pockets between the primary shielding walls, the pockets in the rows being separated from other pockets by the primary shielding walls, the pockets in the columns the separated from other pockets by the secondary shielding walls.
10. The electrical connector of claim 1, wherein the contact holder includes a base and a tower extending from the base, the base having a top and a bottom, the base being received in the pocket, the tower extending from the top of the base to the exterior of the conductive housing, the conductive housing providing circumferential shielding around the base.
11. The electrical connector of claim 10, wherein the contacts extend from the bottom of the base to the circuit board, the conductive housing providing circumferential shielding around the portions of the contacts below the bottom of the base.
12. The electrical connector of claim 1, wherein the conductive housing is manufactured from a plated plastic material, from a conductive polymer, or from a metal material.
13. The electrical connector of claim 1, wherein the circumferential shielding of the conductive housing extends entirely circumferentially around each pocket.
14. The electrical connector of claim 1, wherein the circumferential shielding of the conductive housing extends at least partially circumferentially around each pocket covering greater than 50% of a perimeter of the pocket.
15. An electrical connector comprising: a conductive housing having a mating end and a mounting end, the mounting end configured to be mounted to a circuit board, the conductive housing including shielding walls extending between the mating end and the mounting end, the shielding walls defining shield boxes surrounding pockets, the shield boxes providing electrical shielding on all sides of the pockets, the conductive housing including ground pads along bottoms of the shielding walls configured to be terminated to the circuit board to electrically connect each shielding wall to the circuit board; contact assemblies held by the conductive housing, each contact assembly including a contact holder holding at least one contact, the contact holder being received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing, each contact includes a mating end and a terminating end, the terminating end configured to be coupled to a signal pad of the circuit board; wherein the ground pads provide 360° shielding around the terminating ends of the contacts; and wherein the shield boxes provide circumferential shielding around each contact assembly between the mating end and the mounting end of the conductive housing.
16. The electrical connector of claim 15, wherein the ground pads are coplanar with the terminating ends of the contacts.
17. The electrical connector of claim 15, wherein the ground pads form a grid including ground pad strips and ground pad cross-strips connecting each of the ground pad strips.
18. A communication system comprising: a circuit board including an upper surface having a connector mounting footprint having an array of signal pads arranged in pairs, the signal pads being arranged in columns and rows, the circuit board including a ground grid being electrically connected to a ground plane of the circuit board, the ground grid surrounding the pairs of signal pads, the ground grid including ground strips extending along the columns and cross-strips extending along the rows, the cross-strips connecting each of the ground strips; and an electrical connector mounted to the connector mounting footprint, the electrical connector including a conductive housing and contact assemblies held by the conductive housing, the conductive housing having a mating end and a mounting end mounted to the circuit board, the conductive housing

including pockets extending between the mating end and the mounting end, the conductive housing providing electrical shielding around the pockets, the conductive housing including ground pads at the mounting end terminated to the ground grid of the circuit board, each contact assembly including a contact holder holding a pair of contacts, the contact holder being received in the corresponding pocket of the conductive housing to position the at least one contact relative to the conductive housing, each contact includes a mating end and a terminating end, the terminating end coupled to the corresponding signal pad of the circuit board, wherein the conductive housing provides circumferential shielding around each contact assembly between the contact holder and the ground grid.

19. The communication system of claim 18, wherein the conductive housing includes shielding walls forming the pockets, the shielding walls providing shielding on all sides of the pockets.

20. The communication system of claim 18, wherein the ground pads form a grid including ground pad strips and ground pad cross-strips connecting each of the ground pad strips, the ground pad strips being connected to the ground strips of the ground grid, the ground pad cross-strips being connected to the cross-strips.